

Research papers

Impact of uncertainty on optimal battery operation for price arbitrage and peak shaving: From perspectives of analytical solutions and examples

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ABSTRACT

Uncertain renewable power generation and on-site demand complicate battery operation and affect profitability. In this paper, a novel battery scheduling model (including the single- and multi-interval patterns) is proposed for maximizing arbitrage benefits while meeting the peak shaving requirements. The proposed model features accurate expectations and is essentially a stochastic optimization accounting for uncertainty. The convexity of the proposed model is strictly proved, which ensures that the global optimal solution is obtainable and computationally attractive. Based on the first-order optimality condition, we further derive the analytical form of the optimal solution, which explicitly demonstrates the impact of uncertainty on the optimal schedule of battery storage. Both the convexity and solution stability of the proposed model are verified through case studies involving various time periods and multiple algorithms for convex optimization. In addition, the change in the trade-off price threshold caused by uncertainty is also illustrated in simulations.

1. Introduction

As governments, industries, and investors plan the next stage of energy transformation, the competitiveness of renewable power generation has become increasingly evident. This brings electricity storage, particularly the electrochemical battery, to the center stage [1]. According to recent reports, battery storage in stationary applications is expected to grow at least seventeen-fold between 2017 and 2030 [2]. With this booming prospect, achieving the maximum application benefits of battery storage has raised extensive concerns [3–5].

Battery storage devices are now widely deployed in the generation [6], network [7], and demand sides [8] of power systems. Their role has expanded from overcoming power shortages [9] to improving the system's stability [10], flexibility [11], and economy [12]. Among the various applications of battery storage, price arbitrage [13–15] and peak shaving [16–18] are the two most popular applications in power systems. The former achieves economic benefits by charging batteries with low-priced electricity and discharging them during periods of high prices, whereas the latter provides capacity relief by releasing electricity during peak-load periods.

Regarding the application of price arbitrage, Metz et al. [19] identified the price volatility level that is required to generate a profit from

arbitrage operations, where the battery calendric lifetime and limitations to the cycle number are considered. Liu et al. [20] aggregated distributed battery devices for price arbitrage in the energy market. In [21], the available battery storage systems were coordinated in the presence of renewable energy sources and a heuristic method was proposed to enhance arbitrage benefits under dynamic price signals. In terms of the battery participation in peak shaving, Pholboon et al. [22] presented a real-time battery management algorithm for peak demand shaving in small energy communities, achieving a 35 % reduction in peak energy consumption. Shi et al. [23] proposed a joint optimization framework that includes battery degradation, peak shaving threshold, and uncertainty in customer load. Manojkumar et al. [24] adopted a rule-based peak-shaving strategy with dynamic demand and feed-in limits. Based on metaheuristic and mathematical optimization approaches, a novel battery energy management unit was developed in [25] that minimizes the power consumption of buildings during peak hours. Furthermore, as the stacked application benefits of battery storage are revealed [26], the coordination between price arbitrage and peak shaving using battery storage systems has been extensively studied [27–29]. Hao et al. [30] presented a generalized battery model and developed optimal operation algorithms to provide price arbitrage and peak-shaving services. Schneider et al. [31] proposed a novel battery planning approach that supports the selection of battery technology and

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Nomenclature**Abbreviations**

PV	photovoltaic
SOC	state of charge
PDF	probability distribution function
CDF	cumulative distribution function

Symbols

t	index of time intervals
N	total number of time intervals
T	duration of a time interval
λ_p	price of buying electricity
λ_s	price of selling electricity
λ_b	price of battery degradation
λ_{vp}	virtual purchase price
P	forecast value of net demand
$\tilde{\varepsilon}$	forecast error of net demand
$P_{\min}$	lower limit of net demand
$P_{\max}$	upper limit of net demand
P_{th}	threshold of peak shaving
P_g	transaction power with upstream grid

$P_{b, \max}$	rated power of batteries
$P_b^{\min}$	lower limit of battery power determined by SOC and $P_{b, \max}$
$P_b^{\max}$	upper limit of battery power determined by SOC and $P_{b, \max}$
P_b	battery power
P_b^d	battery discharging power
P_b^c	battery charging power
S_0	battery SOC at initial time
S_t	battery SOC at interval t
S_N	battery SOC at final time
$S_{\min}$	lower limit of battery SOC
$S_{\max}$	upper limit of battery SOC
E_b	rated capacity of batteries
$E(\tilde{c})$	expectation of operational cost
$E(\tilde{c}_1)$	expectation of battery degradation cost
$E(\tilde{c}_2)$	expectation of electricity transaction cost
$f_{\tilde{\varepsilon}}(\cdot)$	probability density function of $\tilde{\varepsilon}$
$f_P(\cdot)$	probability density function of net demand
$F_P(\cdot)$	cumulative distribution function of net demand
$F_P^{-1}(\cdot)$	inverse function of $F_P(\cdot)$

size for shaving and arbitrage. The economic returns of price arbitrage and peak shaving using batteries were demonstrated in [32] based on practical commercial buildings.

However, renewable power generation and electrical load are both random and cannot be accurately predicted, which aggravates the difficulties of optimizing battery operation [33,34]. Hashmi et al. [35] employed a linear programming approach to analyze the impact of uncertain renewable power generation on the operation of battery storage. To control the risk of uncertainty, a fuzzy decision strategy for battery arbitrage was presented in [36]. Cao et al. [37] developed a chance-constrained optimization method for battery storage to maintain uncertain power fluctuations within predefined limits. In addition, the uncertainties associated with renewable power production and loads can be modeled using discrete sample scenarios [38]. Considering the profit of the distribution company, Abdeltawab et al. [39] utilized a scenario-based programming approach to provide localized reactive power support, peak shaving, and price arbitrage by battery storage.

The coordination between price arbitrage and peak shaving enables battery users to improve their economic benefits and grid friendliness. This study aims to optimize the operation of battery storage devices by performing price arbitrage and peak shaving under the uncertainty of the net demand. Net demand denotes the difference between the electrical load and renewable power generation, both of which can be uncertain. With the electrification of end-use energy and the promotion of renewable energy, the impact of uncertainty on optimal battery operation remains to be investigated [40,41]. On the one hand, the existing battery scheduling models that consider uncertainty are similar to black boxes (e.g., [33]), which makes it difficult for operators to distinguish the influence mechanism of uncertainty on optimal battery storage operation. On the other hand, the scenario-based stochastic optimization method is currently one of the mainstream ideas for optimizing the operation of batteries under uncertainty [42–45]. However, the mathematical expectation of the operational costs (i.e., the objective to be optimized) in this class of methods is only an approximation obtained by summing discrete scenarios. Owing to the complex mathematical integrals, few studies have explored the battery scheduling strategy under exact expectations.

To fill the above research gaps, this paper proposes a day-ahead stochastic scheduling model for batteries to simultaneously perform price arbitrage and peak shaving. This model has three remarkable

features. First, the proposed model accurately formulates the expected cost of buying or selling electricity from/to the grid under the uncertainty of the net demand, which is an improvement over existing scenario-based approximations. Second, the analytical optimal solution of the simplified model (considering only a single time interval, Section 2) is derived using the first-order optimality condition. This explicitly shows how the optimal battery schedule decision is affected by the uncertain net demand. Third, the convexity of the proposed model is rigorously proved (Section 3). This is an excellent property from a mathematical optimization point of view, which guarantees that the globally optimal solution is obtainable and computationally attractive.

Compared to other studies, the contributions of this study are twofold. On the one hand, we analyze the impact of uncertain net demand on the optimal battery schedule. The new price threshold (referred to as the virtual purchase price in Section 4) caused by uncertainty is also elaborated in this study, which defines whether it is economically beneficial to enable battery storage systems. To the best of our knowledge, this is the first study to explore the influence mechanism of uncertainty on battery operation in an analytical way. On the other hand, this study provides a new modeling method for the optimal combined operation of price arbitrage and peak shaving for battery storage under uncertainty. That is, to precisely model the cost expectations by means of numerical integration rather than traditional sample approximations. Theoretical analysis and simulation results show that this method is solvable and effective.

The remainder of this paper is organized as follows. Section 2 introduces the formulation of the proposed battery scheduling model in a single-interval situation. The impact of uncertain net demand on the optimal battery schedule is also analyzed in this section. Section 3 further generalizes the battery operating situation into multi-interval (e.g., day-ahead) scheduling and proves the convexity of the proposed model. Both the model performance and impacts of uncertainty on the optimal battery schedule are validated in Section 4. Finally, Section 5 concludes the paper.

2. Impact analysis of uncertainty: based on single-interval battery scheduling model

In this section, a general formulation for the single-interval battery operating decision is established, by which the optimal battery schedule

is determined, considering the uncertainty of the net demand. The resultant problem is proven to be convex. In addition, the optimal solution of the proposed model under various conditions is strictly derived, which explicitly reflects the impact of uncertainty on the optimal battery schedule.

2.1. Representation for uncertainty: probabilistic forecast

Compared with the traditional deterministic point forecast [46], the probabilistic forecast can provide the probability distribution characteristics of forecast errors apart from forecast values, which helps grasp all possible power realizations in the future. Typically, the forecast errors are modeled with a Gaussian distribution [47], beta distribution [48], Cauchy distribution [49], or other nonparametric distributions [50]. Furthermore, Pinson et al. [51] and Zhang et al. [52] proposed detailed modeling methods for the probability distribution function (PDF) of forecast errors from the perspective of empirical approaches and historical statistics, respectively. Overall, error distributions, although possible in various types, are available during battery operation. This study assumes that the point forecasts and PDFs of the forecast errors are known. Without loss of generality, the net demand, that is, the difference between the power demand and local power generation, can be expressed as

$$\tilde{P} = P + \tilde{\varepsilon}, \quad (1)$$

where P denotes the forecast value of net demand. $\tilde{\varepsilon}$ denotes the forecast error, which appears as a random variable in this study and is subject to PDF $f_{\tilde{\varepsilon}}(\cdot)$. Therefore, net demand $\tilde{P}$ is also random, and its PDF is defined as $f_{\tilde{P}}(\cdot)$. The relationship between $f_{\tilde{P}}(\cdot)$ and $f_{\tilde{\varepsilon}}(\cdot)$ can be expressed as $f_{\tilde{P}}(x) = f_{\tilde{\varepsilon}}(x - P)$.

2.2. Single-interval battery scheduling model

In a smart microgrid with a battery storage system, the power supply and demand should always be balanced:

$$P_b + P_g = \tilde{P}, \quad (2)$$

where P_b and P_g denote the battery power and power from/to the upstream grid, respectively. $P_b, P_g > 0$ means that batteries are discharged or the upstream grid supplies power to the microgrid. Conversely, $P_b, P_g < 0$ indicates that the battery or the upstream grid absorbs electricity.

Power transactions with an upstream grid result in electricity bills. The charging and discharging behaviors cause battery degradation. Electricity bills and battery degradation cost constitute the daily operational cost for the smart microgrid, the formulas of which can be written as

$$\begin{aligned} \tilde{c} &= T \cdot [\lambda_p \cdot \max(P_g, 0) + \lambda_s \cdot \min(P_g, 0) + \lambda_b \cdot |P_b|] \\ &= \begin{cases} T \cdot [\lambda_p \cdot (\tilde{P} - P_b) + \lambda_b \cdot |P_b|] & \tilde{P} \geq P_b \\ T \cdot [\lambda_s \cdot (\tilde{P} - P_b) + \lambda_b \cdot |P_b|] & \tilde{P} < P_b \end{cases} \end{aligned} \quad (3)$$

where T denotes the duration of a time interval, and λ_p, λ_s , and λ_b denote the electricity purchase price, sale price, and battery degradation price, respectively.

However, local power generation and demand will occur in the future, lagging battery scheduling. Thus, the objective of minimizing the operational cost is in fact to minimize the expectation of the operational cost, that is,

$$\begin{aligned} \min_{P_b} E(\tilde{c}) &= T \cdot \int_{P_b}^{P_{\max}} \lambda_p \cdot (\tau - P_b) \cdot f_{\tilde{P}}(\tau) d\tau + T \cdot \lambda_b \cdot |P_b| \\ &\quad + T \cdot \int_{P_{\min}}^{P_b} \lambda_s \cdot (\tau - P_b) \cdot f_{\tilde{P}}(\tau) d\tau \end{aligned} \quad (4)$$

$$\text{s.t. } P_b^{\min} \leq P_b \leq P_b^{\max} \quad (5)$$

$$P_{\max} - P_b \leq P_{th}, \quad (6)$$

where $P_b^{\min}$ and $P_b^{\max}$ denote the lower and upper limits of battery power, respectively, which reflect the power constraints imposed by the rated battery power and state of charge (SOC) limits. $P_{\min}$ and $P_{\max}$ denote the lower and upper limits of the net demand, respectively. P_{th} denotes the threshold for peak shaving.

2.3. Optimal solution of single-interval model

As previously mentioned, optimal battery scheduling can be formulated as a problem of minimizing the expected cost. In this subsection, we prove the nature of the proposed model, that is, convexity. The optimal solution in analytical form is derived based on the first-order optimality condition.

For convenience, the objective function (4) is rewritten as:

$$E(\tilde{c}) = E(\tilde{c}_1) + E(\tilde{c}_2), \quad (7)$$

where

$$E(\tilde{c}_1) = T \cdot \lambda_b \cdot |P_b| \quad (8)$$

$$E(\tilde{c}_2) = T \cdot \left[\int_{P_b}^{P_{\max}} \lambda_p \cdot (\tau - P_b) \cdot f_{\tilde{P}}(\tau) d\tau + \int_{P_{\min}}^{P_b} \lambda_s \cdot (\tau - P_b) \cdot f_{\tilde{P}}(\tau) d\tau \right]. \quad (9)$$

On the one hand, because the coefficients λ_b and T in (8) are greater than zero, $E(\tilde{c}_1)$ is an open-up absolute value function and is convex [53]. The first-order derivative is given by Eq. (10). The second-order derivative is always zero. By contrast, $E(\tilde{c}_2)$ is a complex nonlinear function with integral terms. Its first- and second-order derivatives are expressed as (11)–(12).

$$dE(\tilde{c}_1)/dP_b = \begin{cases} \lambda_b \cdot T & P_b > 0 \\ -\lambda_b \cdot T & P_b < 0 \end{cases}, \quad (10)$$

$$dE(\tilde{c}_2)/dP_b = T \cdot [(\lambda_p - \lambda_s) \cdot F_{\tilde{P}}(P_b) - \lambda_p], \quad (11)$$

$$d^2E(\tilde{c}_2)/dP_b^2 = T \cdot (\lambda_p - \lambda_s) \cdot f_{\tilde{P}}(P_b), \quad (12)$$

where $F_{\tilde{P}}(\cdot)$ denotes the cumulative distribution function (CDF) of net demand. Generally, the electricity purchase price λ_p is designed to be no less than the sale price λ_s [54,55]. In addition, $f_{\tilde{P}}(\cdot) \geq 0$, $T > 0$, and (12) ≥ 0 . The nonnegative second-order derivative indicates that $E(\tilde{c}_2)$ is globally convex.

Thus, the objective function $E(\tilde{c})$ is convex according to Theorem 1. In particular, when uncertainty is ignored, $f_{\tilde{P}}(\cdot)$ degenerates into a unit-impulse function while its nonnegative nature is maintained. Therefore, the established operational cost remains a convex function under deterministic conditions.

Theorem 1. A nonnegative weighted sum of convex functions is still convex [53].

Furthermore, based on (10)–(12), the first- and second-order derivatives of the objective function can be written as

$$dE(\tilde{c})/dP_b = T \cdot \begin{cases} (\lambda_p - \lambda_s) \cdot F_{\tilde{P}}(P_b) - \lambda_p + \lambda_b & P_b > 0 \\ (\lambda_p - \lambda_s) \cdot F_{\tilde{P}}(P_b) - \lambda_p - \lambda_b & P_b < 0 \end{cases}, \quad (13)$$

$$d^2E(\tilde{c})/dP_b^2 = T \cdot (\lambda_p - \lambda_s) \cdot f_{\tilde{P}}(P_b) \quad P_b \neq 0. \quad (14)$$

It can be noticed that the first-order derivative of $E(\tilde{c})$ is a piecewise continuous function. Ignoring the limitation of P_b in (5)–(6) temporarily,

$$\lim_{P_b \rightarrow -\infty} \{dE(\tilde{c})/dP_b\} = -\lambda_p - \lambda_b, \quad (15)$$

$$\lim_{P_b \rightarrow 0^-} \{dE(\tilde{c})/dP_b\} = (\lambda_p - \lambda_s) \cdot F_p(0) - \lambda_p - \lambda_b, \quad (16)$$

$$\lim_{P_b \rightarrow 0^+} \{dE(\tilde{c})/dP_b\} = (\lambda_p - \lambda_s) \cdot F_p(0) - \lambda_p + \lambda_b, \quad (17)$$

$$\lim_{P_b \rightarrow +\infty} \{dE(\tilde{c})/dP_b\} = \lambda_b - \lambda_s. \quad (18)$$

$\therefore \lambda_p > 0, \lambda_b > 0, F_p(0) \in [0, 1], \therefore (15) < 0, (16) < 0$. However, (17)–(18) can be positive or negative, depending on the price conditions, which are classified as follows:

Condition 1. $(18) \leq 0$. In this case, $\therefore (14) \geq 0$, and $E(\tilde{c})$ is continuous, $\therefore (17) \leq 0$, and the objective function monotonically decreases globally (if $(17) \neq 0$) or is at least a generalized decline (if $(17) = 0$). Considering the limitation of P_b in (5)–(6), the global optimum of the proposed model is taken at the upper limit of the definition domain, that is,

$$P_b^* = P_b^{\max}. \quad (19)$$

Condition 2. $(18) > 0, (17) < 0$. In this case, $\therefore (14) \geq 0$, and $dE(\tilde{c})/dP_b$ is continuous when P_b is greater than zero, $\therefore \exists P^* \in (0, +\infty)$ holds $dE(\tilde{c})/dP_b|_{P^*} = 0$:

$$(\lambda_p - \lambda_s) \cdot F_p(P^*) - \lambda_p + \lambda_b = 0, \quad (20)$$

$$P^* = F_p^{-1}[(\lambda_p - \lambda_b)/(\lambda_p - \lambda_s)]. \quad (21)$$

In addition, $\therefore E(\tilde{c})$ is continuous, $\therefore$ the objective function monotonically decreases in $(-\infty, P^*)$, and increases in $(P^*, +\infty)$. Thus, within the original feasible regions (5)–(6), the global optimum of the proposed model is:

$$P_b^* = \begin{cases} P_b^{\max} & P_b^{\max} \leq P^* \\ P^* & P_b^{\min*} < P^* < P_b^{\max} \\ P_b^{\min*} & P^* < P_b^{\min*} \end{cases}, \quad (22)$$

where

$$P_b^{\min*} = \max(P_b^{\min}, P_{\max} - P_{th}). \quad (23)$$

Condition 3. $(18) > 0, (17) \geq 0$. In this case, $\therefore (14) \geq 0$, and $E(\tilde{c})$ is continuous, $\therefore$ the objective function monotonically decreases in $(-\infty, 0)$, and increases in $(0, +\infty)$. Considering the limitations of P_b in (5)–(6), the global optimum of the proposed model is:

$$P_b^* = \begin{cases} P_b^{\max} & P_b^{\max} \leq 0 \\ 0 & P_b^{\min*} < 0 < P_b^{\max} \\ P_b^{\min*} & 0 < P_b^{\min*} \end{cases}. \quad (24)$$

2.4. Impact of uncertainty on optimal solution

Eqs. (19) and (22)–(24) show the optimal battery schedule with analytical forms considering uncertain net demand. Similarly, the optimal battery schedule without uncertainty can be derived as:

Condition 1

$$P_b^* = P_b^{\max}, \quad (25)$$

Condition 2

$$P_b^* = \begin{cases} P_b^{\max} & P_b^{\max} \leq P \\ P & P_b^{\min*} < P < P_b^{\max} \\ P_b^{\min*} & P < P_b^{\min*} \end{cases}, \quad (26)$$

Condition 3

$$P_b^* = \begin{cases} P_b^{\max} & P_b^{\max} \leq 0 \\ 0 & P_b^{\min*} < 0 < P_b^{\max} \\ P_b^{\min*} & 0 < P_b^{\min*} \end{cases}. \quad (27)$$

By comparison, it can be seen that uncertainty affects the optimal battery schedule only under **Condition 2**. Specifically, if the forecast is completely accurate, the operator can determine the optimal battery schedule based only on the predicted net demand P . However, in the scenario with uncertainty, the key that affects the battery operating decision is shifted from P to a new value, P^* , as defined in (21). Additionally, the uncertainty of the net demand would change the price threshold that serves as the dividing line of battery profitability, which is illustrated in **Section 4.1**.

Note that $F_p^{-1}(\cdot)$ in (21) is the inverse function of the corresponding CDF and is sometimes difficult to express explicitly. In this case, the Newton-Raphson method is recommended for obtaining the numerical solution owing to its simplicity and fast convergence [56]. In other cases (**Conditions 1 and 3**), the optimal battery schedule can be quickly solved only by simple logical judgments.

3. Optimal daily operation: based on multi-interval battery scheduling model

In the previous section, we present a single-interval battery scheduling model. The convexity and optimal solution of the proposed model are derived in detail. In this section, the above model is generalized to the daily scheduling scenario with multiple time intervals.

3.1. Multi-interval battery scheduling model

In a scenario with multiple time intervals, the optimization objective is to minimize the expectation of the sum of the operational cost in each interval, that is,

$$\begin{aligned} \min_{b,t} E\left(\sum_{t=1}^N \tilde{c}_t\right) &= \sum_{t=1}^N E(\tilde{c}_t) \\ &= \sum_{t=1}^N \lambda_b \cdot |P_{b,t}| \cdot T \\ &+ \sum_{t=1}^N T \cdot \left[\int_{P_{b,t}}^{P_{\max}} \lambda_{p,t} \cdot (\tau - P_{b,t}) \cdot f_{p,t}(\tau) d\tau \right] \\ &+ \sum_{t=1}^N T \cdot \left[\int_{P_{\min}}^{P_{b,t}} \lambda_{s,t} \cdot (\tau - P_{b,t}) \cdot f_{p,t}(\tau) d\tau \right] \end{aligned} \quad (28)$$

where t and N denote the interval index and total number of intervals, respectively. The equation for Line 1 is based on **Theorem 2**. Line 2 denotes the degradation cost of batteries, which is deterministic and piecewise linear. Lines 3 and 4 reflect the expected electricity transaction cost, which consists of the expected electricity purchase cost (Line 3) and expected electricity sale revenue (Line 4). The expected electricity transaction cost is continuous and nonlinear.

Theorem 2. The expected value of the sum of multiple functions is the sum of the expected values of the functions [57].

The operational constraints of batteries are stated as (29)–(35).

$$-P_{b,\max} \leq P_{b,t} \leq P_{b,\max} \quad (29)$$

$$P_{t,max} - P_{b,t} \leq P_{th} \quad (30)$$

$$P_{b,t} = P_{b,t}^d - P_{b,t}^c \quad (31)$$

$$S_t = S_{t-1} - \left(P_{b,t}^d \eta^d - P_{b,t}^c \eta^c \right) \cdot T / E_b \quad (32)$$

$$S_{\min} \leq S_t \leq S_{\max} \quad (33)$$

$$S_0 = S_N \quad (34)$$

$$P_{b,t}^d \cdot P_{b,t}^c = 0, \quad (35)$$

where $P_{b, \max}$ and E_b denote the rated power and rated capacity of batteries, respectively. $P_{b,t}^d$ and $P_{b,t}^c$ denote the battery discharging power and charging power, respectively. S_0 , S_b and S_N denote the SOC at different times, that is, the initial time, interval t , and final time, respectively. $S_{\min}$ and $S_{\max}$ denote the lower and upper limits of battery SOC, respectively.

3.2. Analysis and solution of multi-interval model

As can be seen from line 1 of (28), the objective function of the multi-interval model is transformed into the sum of the expected operational costs for every interval. The convexity of every addend (i.e., the expected cost of interval t) is proved in Section 2.3. Therefore, the objective function of the multi-interval model is still convex according to Theorem 1.

Eq. (35) reflects the constraint on the charging and discharging status of batteries, which can be relaxed without affecting the accuracy of the model [58]. The other constraints have clear linear features. Constraints (29)–(31) can be combined into one, as expressed in (36). Constraints (32) and (33) are combined as (37). Likewise, constraint (34) in combination with (32) is simplified to (38).

$$\max(-P_{b,max}, P_{t,max} - P_{th}) \leq P_{b,t}^d - P_{b,t}^c \leq P_{b,max} \quad (36)$$

$$(S_0 - S_{\max}) \cdot E_b / T \leq \sum_{t=1}^i (P_{b,t}^d \eta^d - P_{b,t}^c \eta^c) \leq (S_0 - S_{\min}) \cdot E_b / T \quad i = 1 \dots N \quad (37)$$

$$\sum_{t=1}^N (P_{b,t}^d \eta^d - P_{b,t}^c \eta^c) = 0 \quad (38)$$

The resultant constraints, that is, (36)–(37) and (38), formulate polyhedral sets and a hyperplane, respectively, both of which are convex [53]. Thus, the feasible domain of the proposed model is a convex set according to Theorem 3.

Theorem 3. The convexity is preserved under the intersection of an infinite number of convex sets [53].

In conclusion, the multi-interval optimal battery scheduling model is a convex programming problem that can be effectively solved using advanced algorithms.

4. Case study

In this section, we validate the proposed model from multiple timescales (single interval, two intervals, and one day) through numerical examples. The single-interval case illustrates the stable convex shape of the proposed model under various probabilistic distributions and price conditions. The impact of uncertainty on the dividing price line of the battery profitability is also shown in this case. The dual-interval scenario visualizes the convexity of the multi-interval model. Finally, the day-ahead case verifies the household practicability of the proposed model.

One-day residential power profiles of PV generation and loads from the public dataset *Smart** [59] are employed in this section. The installed capacities of the PV cells and loads were fixed at 0.5 kW and 0.8 kW, respectively. The peak-shaving threshold was set to 2 kW. We assume that the residential user is equipped with *Tesla's Powerwall*, whose detailed parameters are listed in Table 1. Additionally, considering the various practices in residential electricity prices around the world, two representative price levels, that is, China (for the single- and dual-interval cases) and the United States (for the day-ahead case), are referred to for the test.

4.1. Case 1: single-interval battery scheduling

The uncertainty of the PV power and loads, which can be expressed with the PDFs of forecast errors, significantly influences the optimal battery schedule. In this subsection, several typical distribution forms mentioned above, that is, Gaussian, beta, and Cauchy, are included to investigate the performance of the single-interval model. Three sets of rate levels derived from residents' TOU tariffs [60] in Shanghai, China are also involved.

The detailed case parameters are listed in Table 2. The battery SOC was initially set at 70 %. The configuration principle of the probability distribution parameters is to ensure that the peak values of the PDFs are typically obtained when the forecast errors are zero. At the same time, to verify that the convexity of the proposed model is independent of the shape of the default distributions, the probability distribution parameters are calibrated to include as many features as possible, for example, symmetric/asymmetric, unbiased/left-bias/right-bias.

The overall shapes of the expected operational cost curves are shown in Fig. 1. The magenta plus signs indicate the optima. One can notice that although the expected cost varies with the probability distribution type, their convex trends are consistent and perfectly agree with our theoretical derivation in Section 2.3.

We abbreviate $\lambda_p - (\lambda_p - \lambda_s)F_p(0)$ (the opposite of the first two terms in (16)) as λ_{vp} and name it the virtual purchase price. Obviously, $\lambda_s \leq \lambda_{vp} \leq \lambda_p$. Further, as analyzed in Section 2.3, there are three possible model evolutions depending on the different price orders (by size): $\lambda_b \leq \lambda_s \leq \lambda_{vp}$ (corresponding to Condition 1), $\lambda_s < \lambda_b < \lambda_{vp}$ (corresponding to Condition 2), and $\lambda_s \leq \lambda_{vp} \leq \lambda_b$ (corresponding to Condition 3). When the battery cost λ_b is not higher than the electricity sale price λ_s (during this time, discharging batteries is profitable), the more electricity the battery discharges, the more the user benefits. These benefits come from the reduction in buying electricity or an increase in selling electricity (Condition 1, Fig. 1(a)). As the battery cost λ_b rises and exceeds the electricity sale price λ_s , excessive discharge implies spending the high-value battery loss in exchange for the low-value sale electricity profits, and insufficient discharge means that more power will be bought at a high price from the grid. Both lead to increased expected costs. Thus, in this case, there is a tradeoff between the high and low battery discharge powers, that is, the optimum under Condition 2 (Fig. 1(b)). When the battery cost λ_b gradually approaches or even exceeds the electricity purchase price λ_p , it is not profitable to use the battery. Thus, the optimal battery power is zero (Condition 3, Fig. 1(c)). In fact, in light of our derivation of the optimal battery schedule under uncertainty in Section 2.3, as long as λ_b is higher than the virtual purchase price λ_{vp} (not necessarily higher than the real purchase price λ_p), battery operation is

Table 1
Basic parameters of Tesla's Powerwall.

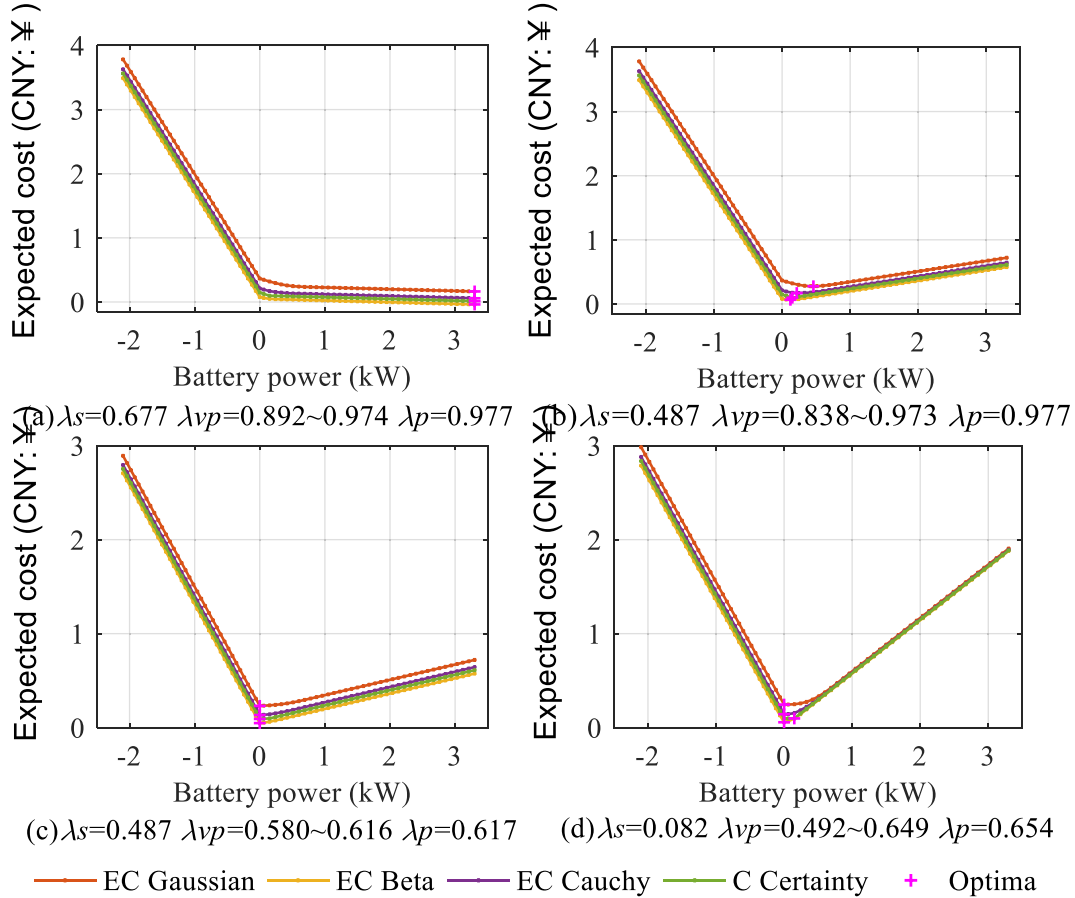
$E_{b, \max}$ (kWh)	$P_{b, \max}$ (kW)	$SOC_{\min}$ (%)	$SOC_{\max}$ (%)	INV^a (\$)	WAR^a (years)	$\eta^c, 1/\eta^d$
7	3.3	15	100	5000	10	0.95

^a INV denotes the total investment costs. WAR denotes the warranty period. Thus, the battery's degradation cost is estimated as $\lambda_b = 0.098\$/kWh$ (0.65 ¥/kWh) according to the method provided by [23].

Table 2

Configuration parameters of single-interval case.

Model parameter	P_{pv} (kW)/ P_l (kW)	λ_p (¥/kWh)	λ_s (¥/kWh)
	0.25/0.4	0.977/0.977/0.617	0.677/0.487/0.487
Probability distribution parameter ^a	Gaussian $\tilde{e}_{pv} \sim G(0, 125^2)$ $\tilde{e}_l \sim G(0, 200^2)$ $F_p(0): 0.282$	Cauchy $0.1\tilde{e}_{pv} \sim C(2, 0)$ $0.05\tilde{e}_l \sim C(2, 0)$ $F_p(0): 0.024$	Beta $0.0032\tilde{e}_{pv} + 0.2 \sim B(2, 5)$ $0.002\tilde{e}_l + 0.8 \sim B(5, 2)$ $F_p(0): 0.210$

^a $\tilde{e}_{pv}$ and $\tilde{e}_l$ denote the forecast errors of the local PV generation and demand, which are measured in Watts (W).**Fig. 1.** Expected cost of single-interval case under different conditions.

no longer profitable (the optimal solution is zero), as shown in Fig. 1(d). However, if the point forecast is completely accurate (green curve in Fig. 1(d)), discharging the battery is still cost-effective compared with buying electricity from the grid at the price λ_p . In other words, uncertainty causes λ_{vp} (instead of the purchase price λ_p under deterministic conditions) to take the demarcation role of judging whether it is profitable or not to run the battery. This is also why we name λ_{vp} the virtual purchase price.

4.2. Case 2: dual-interval battery scheduling

As a pioneer of subsequent multi-interval scheduling, in this subsection, a dual-interval case is designed to visually show the convexity of the proposed model.

Since the stable convexity of the model under different probability distributions has been demonstrated in the previous case, it is assumed that the forecast errors satisfy the Gaussian distribution for simplicity. Certainly, other types of probability distribution are applicable to this

model. The charging and discharging efficiency of the battery is set to 1 in this case. The other parameters are listed in Table 3. Fig. 2 shows the overall shape of the expected operational cost when the battery powers at these two intervals are determined simultaneously. Four pairs of dashed lines, that is, cyan, blue, magenta, and green, represent constraints (36)–(37) of the two intervals. The yellow block area surrounded

Table 3Configuration parameters of dual-interval case.^a

First interval	$P_{pv,1}$ (kW)	$\tilde{e}_{pv,1}$	$\lambda_{p,1}$ (¥/kWh)
	0.35	Gaussian (0,50 ²)	0.617
	$P_{l,1}$ (kW)	$\tilde{e}_{l,1}$	$\lambda_{s,1}$ (¥/kWh)
	0.3	Gaussian (0,80 ²)	0.487
Second interval	$P_{pv,2}$ (kW)	$\tilde{e}_{pv,2}$	$\lambda_{p,2}$ (¥/kWh)
	0.3	Gaussian (0,60 ²)	0.977
	$P_{l,2}$ (kW)	$\tilde{e}_{l,2}$	$\lambda_{s,2}$ (¥/kWh)
	0.4	Gaussian (0,120 ²)	0.487

^a $\tilde{e}_{pv}$ and $\tilde{e}_l$ denote the forecast errors of the local PV generation and demand, which are measured in Watts (W).

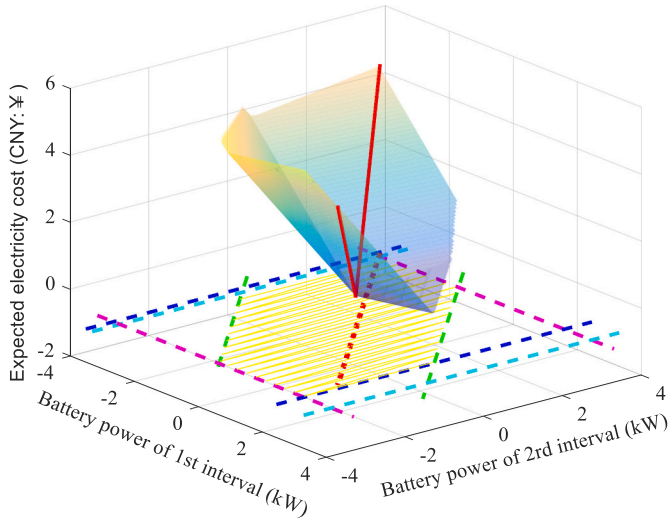


Fig. 2. Expected operational cost of dual-interval case.

by dashed lines denotes the feasible domain of the proposed model, which is the combination of all possible battery powers during the two intervals. From this figure, it can be observed that when the proposed model is extended to the dual-interval scenario from the previous single-interval scenario, both the feasible domain and objective function are still convex. Note that in this case, the constraint (38) that the SOC at the end of the scheduled period should be consistent with the initial SOC is neglected. This is acceptable unless the operational tasks are inherently cyclical (for example, in a one-day operation, operators prefer to return the battery SOC at the final interval to its initial value in preparation for the operation on the next day). Furthermore, if constraint (38) is considered, the feasible domain degenerates to a straight line, that is, the red dash-dotted line in Fig. 2. Accordingly, all the values of the expected operational cost will shrink to a line, that is, the red solid curve in Fig. 2. However, it can be observed that the proposed model is still a convex optimization problem.

Fig. 3 shows the optimization path and iteration process obtained using the interior-point method. It can be noticed that even if the initial point is located at a distant position, it can still effectively (28 iterations) converge to the global optimal solution, i.e., $P_{b,1}^* = 0$ kW, $P_{b,2}^* = 0.142$ kW.

4.3. Case 3: day-ahead battery scheduling

With the performance of the proposed model illustrated in the single- and dual-interval cases, in this subsection, the application scenario of

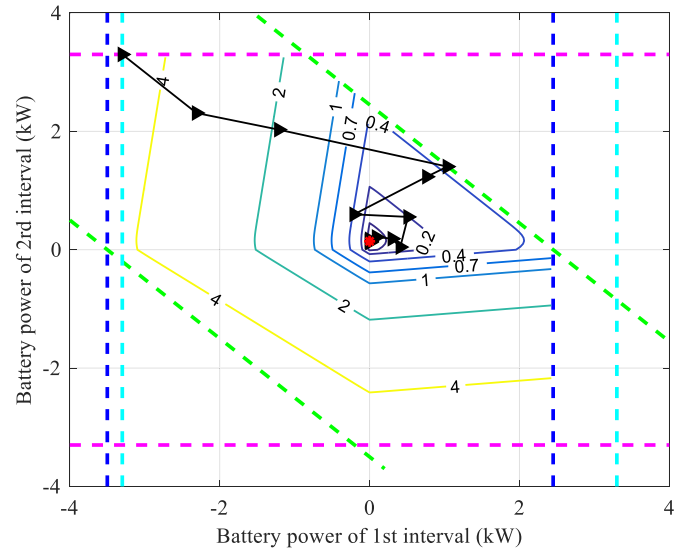


Fig. 3. Locus and process of optimization.

the proposed model is further generalized to day-ahead battery scheduling.

In this case, the one-day metering data of an apartment equipped with the rooftop PV system in *Smart** plan [59], are used as the predicted PV power, as shown in Fig. 4(a). The forecast errors are still assumed to follow a Gaussian distribution, as in Case 2. We set the maximum root mean square errors (RMSE) of the PV power and load forecast to be 20 % and 23.5 %, respectively. In terms of electricity prices, the residential TOU pricing plan [61] from San Diego Gas & Electric Company in California, USA, was adopted, as shown in Fig. 4(b).

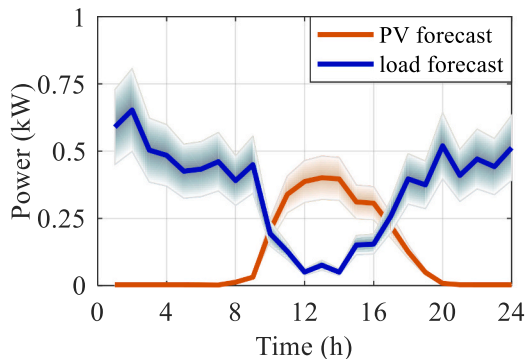
For comparison, three numerical optimization algorithms, that is, the interior-point method, active-set method, and sequential-quadratic-programming (SQP) method, are employed in this subsection. The results are presented in Table 4.

Table 4

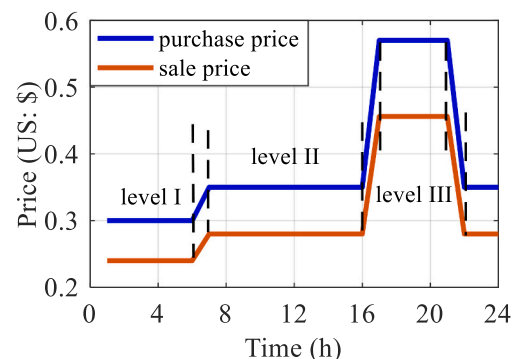
Algorithms comparison about day-ahead case.

Algorithms	Initial	Value	Iterations/generations	Times ^a
Interior point	[0, ..., 0]	\$2.25	314	288 s
Active set	[0, ..., 0]	\$2.25	1171	975 s
SQP	[0, ..., 0]	\$2.25	92	159 s

^a The simulation environment was based on a personal laptop with an Intel i5 3.20GHz CPU, 8G RAM. In the case of the six-core team server with Intel Xeon E5-2620 16G RAM, the computation time is further reduced by at least 30 %.



(a) Probabilistic forecast of PV and load



(b) Electricity purchase and sale prices

Fig. 4. Residential forecast power and TOU prices.

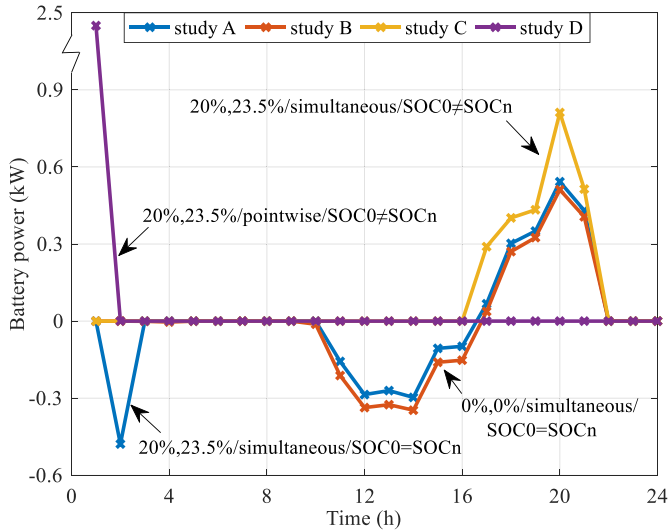


Fig. 5. Optimal battery power schedule in different studies.

For the first three numerical algorithms, the initial points are chosen at $[0, \dots, 0]$, which means that in every time interval, the battery does not work and the user only trades power with the upstream grid to supply the load. Evidently, this initial point satisfies all constraints and must be in the feasible domain. It can be observed from the simulation results that although the solving mechanism is different, all the algorithms converge to the same optimal value, which demonstrates the excellent convergence stability of the proposed model.

The optimal battery power schedule is shown in Fig. 5 (study A). For comparison, we conduct two additional case studies in which the uncertainty of the net demand (study B) and constraint (34) (study C) are separately ignored. The optimal expected operational costs of the three studies are US\$ 2.25, US\$ 2.23, and US\$ 1.30, respectively.

Clearly, study C has the best economic advantage because there is no strict constraint on the battery SOC at the final interval. It can be seen that the high-price zone (17:00–21:00, level III in Fig. 4(b)) is the best period for the arbitrage with battery storage. During this period, the battery is preferentially discharged until it reaches the lowest SOC threshold (Fig. 6). Additionally, the optimal battery power schedule of study C implies that it is not economically worthwhile to charge the battery during the other periods because of the relatively high battery degradation cost (0.2\$/kWh including charging and discharging). Even if the battery is charged at the lowest price (0.3\$/kWh), the resultant cost (0.5\$/kWh) is still higher than the sale price at any time of the day. Instead of incurring battery degradation cost by charging the battery, PV power is more profitable for direct supply to loads or the upstream grid.

If the constraint on the final battery SOC is included, that is, study A, the battery not only reduces the amount of discharged electricity (compared with 17:00–21:00 in study C) but also needs to be charged when the electricity price is lower (2:00 and 11:00–16:00 in study A) to compensate for the SOC.

Additionally, when and how much electricity is needed to charge the battery varies with the uncertainty of the power generation and demand. According to study A and B, it can be observed that the battery system is preferentially charged during the mid-price period (level II) rather than during the low-price level I. This is because the PV power is almost at its peak during mid-day level II, and the charging cost is lower (because the PV generation cost is neglected) than that of level I. On the other hand, uncertain factors have an impact on the optimal battery power schedule. Fig. 5 shows that study A has an additional battery charging process during level I to meet the limit of the final battery SOC. Accordingly, the total operational cost increases by about 1 % from US\$ 2.23 (study B) to US\$ 2.25 (study A) as a response to uncertain conditions. However, completely accurate predictions are not possible in current practice,

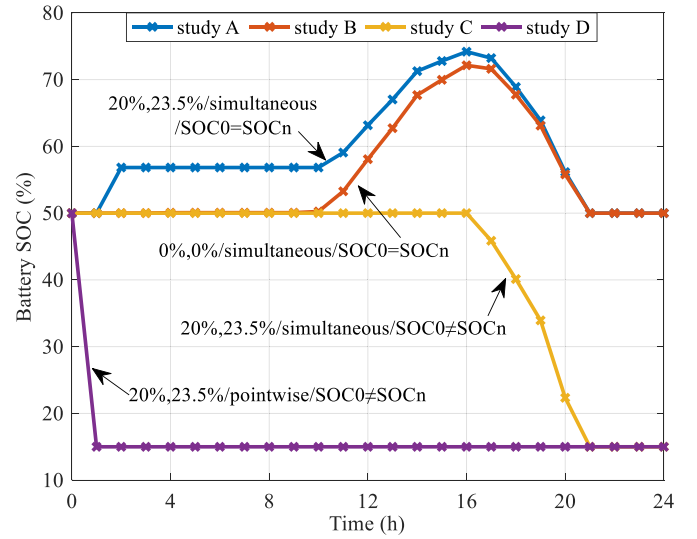


Fig. 6. Battery SOC in different studies.

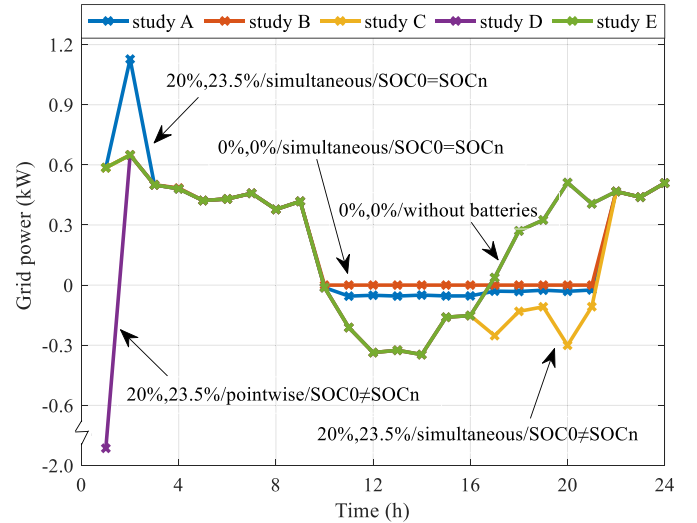


Fig. 7. Expected grid transaction power in different studies.

especially on the residential side, where prediction errors are generally larger than those at the system level. Therefore, it is necessary to consider the uncertainty in optimizing the daily operational costs.

Finally, the same problem is solved by repeating the proposed single-interval model pointwise, that is, study D. It can be seen that the pointwise scheduling mechanism is shortsighted and focuses only on the optimal operational cost at the current time interval. After the electricity in the battery is run out in the first interval, the battery stops working because recharging would result in a degradation cost. In the multi-interval joint scheduling model, the SOC constraints combine the decisions of each interval to provide the model with a global view. As a result, if a resident arranges the battery operation directly in accordance with the pointwise scheme (study D), the expected operational cost will be US\$ 1.99, an increase of approximately 53 % over study C.

Fig. 7 shows the expected electricity transactions between the resident and the grid. Compared with the case without storage (study E), the self-consumption of solar generation in the PV-battery residence significantly increases (10:00–16:00). With the guidance of the proposed model (study A), the residential electricity purchase during the high-price periods (17:00–21:00) is reduced. Further, if the constraint on the final battery SOC is lifted (study C), the battery user can sell the electricity in exchange for more profits.

4.4. Discussion

The economic benefits from price arbitrage and peak shaving are of primary concern to battery users. A discussion of the obtained results summarizing the impact of the uncertain net demand on optimal economic battery operation is provided here.

The battery degradation cost determines whether it is profitable to operate a battery storage device. In the case of high battery degradation costs (for example, *study C*), the electricity generated by PV panels is better sold to the grid directly instead of being charged into the battery and then released in another time interval.

Besides the battery degradation cost, electricity purchase prices affect the optimal economic battery operation. If the electricity purchase price is lower than the battery degradation cost, purchasing power directly from the grid will achieve a lower operational cost compared with using batteries. However, this statement is based on the deterministic scenario. According to this study, under the uncertainty of the net load, the virtual purchase price instead becomes the significant factor that determines the profitability of battery storage. We derive the mathematical expression of the virtual purchase price, which consists of the electricity purchase price, sale price, and cumulative distribution function describing uncertainty.

The above discussions do not include the cost of the electricity in the battery. The stored electricity in the battery may come from PV generation or the grid at the cost of zero or an electricity purchase price, respectively. If this electricity cost is considered, the dividing price line of the battery profitability will be the difference between the virtual purchase price and the cost of the electricity in the battery, not just the virtual purchase price. From a practical point of view, the difference between the highest and lowest electricity purchase prices was previously identified by practitioners as a price threshold to judge the battery profitability, which is based on the deterministic scenario. However, under the uncertainty of the net demand, this price threshold should be adjusted to the difference between the virtual purchase price during the highest-price period and the lowest electricity purchase price. In addition, it is necessary to consider the uncertainty of the net demand in optimizing the daily battery operation because the PV power and loads cannot be predicted accurately.

5. Conclusion

This study proposes an expectation-based stochastic model for optimal battery scheduling in simultaneous price arbitrage and peak shaving. The convexity of the proposed model is strictly proven whether it is for single-interval or multi-interval scheduling and whether the uncertainty is considered, which ensures attractive computation even at the laptop configuration level. The analytical derivation and case study found that the uncertain net load not only affects the optimal battery power but also inherently changes the dividing price line of the battery profitability. Relevant conclusions have significant implications for policy makers to better understand the operation preferences of battery owners and formulate price policies suitable for the development of battery storage. The optimal power curve obtained by the multi-interval model can be used directly by operators to schedule their battery storage systems. In addition, this study does not have strong assumptions about the distribution shapes of forecast errors, which further enhances the practicality and adaptability of the proposed model.

This paper uses the fixed time-of-use price and is mainly oriented to day-ahead battery scheduling. Future work will consider the uncertainty of electricity prices and further extend the proposed scheduling method to intraday battery operation.

CRedit authorship contribution statement

Hao Su: Conceptualization, Formal analysis, Investigation, Methodology, Validation, and Writing.

Donghan Feng: Conceptualization, Methodology, Funding acquisition, Project administration, and Supervision.

Yun Zhou: Formal analysis, Validation, Funding acquisition, and Project administration.

Xing Hao: Investigation, Resources, and Visualization.

Yin Yi: Visualization and Writing.

Kele Li: Resources and Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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